WORKED SOLUTION

STEP I 2015, Question 7

Pure • Maxima on an Interval

STEP I 2015, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q7

The key idea

The greatest value of a smooth function on a closed interval is attained either at a stationary point inside the interval or at an endpoint. Here the stationary points are at $x = 0$ and $x = \frac{a}{3}$, and the second one moves to the right as a grows, eventually leaving the interval. So we list the candidates, then compare them as a varies: the three ranges come from where the comparisons switch and from where $\frac{a}{3}$ passes $x = 1$.

Solution

The candidates

Differentiating,

$$f'(x) = 6ax - 18x^2 = 6x(a - 3x),$$

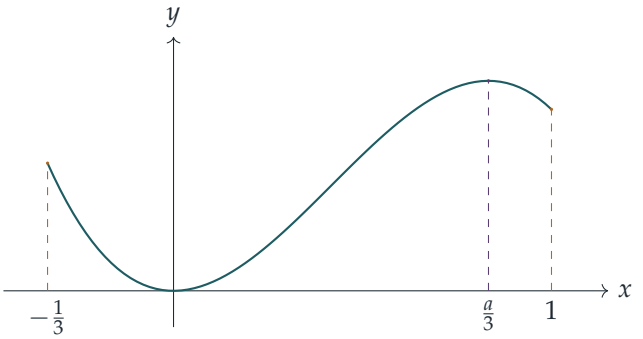
so the stationary points are $x = 0$ and $x = \frac{a}{3}$. For $a \geq 0$ the sign of f' tells us the shape:

- for $x < 0$, both $6x < 0$ and $a - 3x > 0$, so $f'(x) < 0$ and f is decreasing;
- for $0 < x < \frac{a}{3}$, $f'(x) > 0$ and f is increasing;
- for $x > \frac{a}{3}$, $f'(x) < 0$ and f is decreasing.

So $x = 0$ is a local minimum and $x = \frac{a}{3}$ a local maximum (when $a = 0$ the two coincide and $f = -6x^3$ is decreasing throughout). The values we need are

$$f\left(-\tfrac{1}{3}\right) = \frac{a}{3} + \frac{2}{9}, \qquad f\left(\tfrac{a}{3}\right) = \frac{a^3}{3} - \frac{2a^3}{9} = \frac{a^3}{9}, \qquad f(1) = 3a - 6.$$

Because f decreases on $[-\frac{1}{3}, 0]$, the largest value on that piece is $f(-\frac{1}{3})$; the value $f(0) = 0$ is never the greatest. So $\mathrm{M}(a)$ is the largest of $f(-\frac{1}{3})$ and the greatest value of f on $[0, 1]$.



The sketch shows $a = \frac{5}{2}$: the three candidates are the two endpoints (amber) and the local maximum (plum), which here is the winner.

Case $0 \le a \le 3$

Now $\frac{a}{3} \in [0,1]$, so on $[0,1]$ the function rises to $x = \frac{a}{3}$ and then falls. Its greatest value there is $f(\frac{a}{3}) = \frac{a^3}{9}$, and we must compare this with $f(-\frac{1}{3})$:

$$\frac{a^3}{9} - \left(\frac{a}{3} + \frac{2}{9}\right) = \frac{a^3 - 3a - 2}{9} = \frac{(a+1)^2(a-2)}{9}.$$

Ask yourself How do we spot the factorisation of $a^3 - 3a - 2$?

Try small integers: $a = -1$ gives $-1 + 3 - 2 = 0$ and $a = 2$ gives $8 - 6 - 2 = 0$. The sum of the three roots is 0 (there is no a^2 term), so the third root is -1 again, giving $(a+1)^2(a-2)$.

Since $(a+1)^2 > 0$ for $a \ge 0$, the difference has the sign of $a - 2$. Hence

$$\mathrm{M}(a) = \frac{a}{3} + \frac{2}{9} \quad \text{for } 0 \le a \le 2, \qquad \mathrm{M}(a) = \frac{a^3}{9} \quad \text{for } 2 \le a \le 3,$$

with both formulas giving $\frac{8}{9}$ at $a = 2$.

Case $a > 3$

Now $\frac{a}{3} > 1$, so f is increasing on all of $[0,1]$ and its greatest value there is $f(1) = 3a - 6$. Comparing with the left endpoint,

$$(3a - 6) - \left(\frac{a}{3} + \frac{2}{9}\right) = \frac{8a}{3} - \frac{56}{9} = \frac{8}{9}(3a - 7),$$

which is positive since $a > 3 > \frac{7}{3}$. So $\mathrm{M}(a) = 3a - 6$ for $a \ge 3$; at $a = 3$ this is $3 = \frac{27}{9}$, matching the middle formula.

Conclusion

$$\mathrm{M}(a) = \begin{cases} \dfrac{a}{3} + \dfrac{2}{9}, & 0 \le a \le 2, \\ \dfrac{a^3}{9}, & 2 \le a \le 3, \\ 3a - 6, & a \ge 3. \end{cases}$$

In words: for small a the left endpoint wins, for moderate a the local maximum wins, and once the local maximum has left the interval the right endpoint wins.

Checks

- M is continuous: $\frac{8}{9}$ at $a = 2$ from both sides and 3 at $a = 3$ from both sides, as it must be, since the greatest value changes continuously with a.
- $a = 0$: $f(x) = -6x^3$ is decreasing, so its greatest value is $f(-\frac{1}{3}) = \frac{2}{9}$, agreeing with the first formula.
- $a = \frac{5}{2}$: the formula gives $\frac{125}{72} \approx 1.736$, while the endpoints give $f(-\frac{1}{3}) \approx 1.056$ and $f(1) = 1.5$, both smaller. A computer search over a fine grid of x, for several hundred values of a between 0 and 12, agreed with the formula to within 10^{-8}.

Common slips

- Forgetting the left endpoint $x = -\frac{1}{3}$. Because f decreases for $x < 0$, this endpoint can beat the local maximum, and it does for $a < 2$.
- Using $f(\frac{a}{3})$ for $a > 3$, when $\frac{a}{3}$ is outside the interval and so not available.
- Assuming that a local maximum is automatically the greatest value on the interval.

Beyond STEP

Here $\mathrm{M}(a) = \max_{x \in [-1/3, 1]} f(x)$ is a function defined by optimising over a family, and such functions are typically continuous but only piecewise smooth: you might check that the left and right derivatives of M disagree at $a = 2$. That a continuous function on a closed bounded interval attains its maximum at all is the Extreme Value Theorem, proved in a first analysis course; see Abbott, *Understanding Analysis*.

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